

# Owen Yao

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## EDUCATION

### Purdue University

Bachelor of Science, Computer Science

West Lafayette, IN

## EXPERIENCE

### Purdue STARS (Summer Training on Awareness and Readiness for Semiconductors)

West Lafayette, IN

Digital Design & Verification Intern

June - Aug 2025

- Collaborated in a 5 person design team to prototype and tape out a RISC-V CPU graphing-calculator SoC; integrated SPI LCD/keypad; implemented SystemVerilog RTL and programmed using RISC-V assembly.
- Digital design projects: Built FSM-based systems—Rock-Paper-Scissors, Simon Says, Vending Machine—in SystemVerilog; authored unit/system-level testbenches and assertions.

### Lenovo Inc

Morrisville, NC

SDE Intern

June - Aug 2024

- Built a Python tool to automate MI data analysis and chart generation for use by Geo C-level leadership team in weekly reviews, reducing turnaround time by 93%.
- Created proof-of-concept Large Language Models (LLMs) demos and implemented inference tasks to evaluate table Question Answering (Table QA) capability of top table-trained LLMs including TableLlama, TableGPT2, and ChatGPT to guide future dev plan for enabling query of market data.

### Atom Bioworks Inc

RTP, NC

MLOps Intern

June - Aug 2024

- Contributed to AptABLE, an NSF SBIR-funded research project, by building ML pipeline tools for developing a 150M parameters protein language model-based binder classifier and generator, whose accuracy outperformed AlphaFold 3.
- Led the evaluation of MLOps monitoring tools by integrating TensorBoard and Wandb into the ML training workflow, conducting a cost-benefit analysis to recommend Wandb to the MLOps team.

### Department of Material Science Engineering (MSE), NC State University

Raleigh, NC

SDE Intern

June - Aug 2023

- Worked with NCSU undergraduates and postgraduates at the NCSU MSE lab to investigate the ability of memristors to solve non-linear problems by analyzing simulations of memristive networks and circuits.
- Researched effect of voids on percolation; generated void representatives on a network simulation [Percolator and CircuitSymphony] and visualized data using 3D graphical analysis.

## PUBLICATIONS

- Patel, Sawan. Peng, Zhangzhi, Friedman, Adam, Yao, Owen. Fraser, Keith. Chatterjee, Pranam. AptABLE As A Deep Learning Platform for SELEX Optimization. Presented poster at “AI for New Drug Modalities” workshop at NeurIPS 2024.
- Frick, Nikolai. LaBean, Thomas. Trost, Isaac. Yao, Owen. Exploration of Three Dimensional Neuromemristive Material. Presented at: NC State Undergraduate Research and Creativity Summer Symposium. July 27th, 2023. D. H. Hill Jr. Library.

## SKILLS

- **IC design:** SystemVerilog, GTKWave, logic analyzer/oscilloscope, breadboarding
- **Software languages:** Python, Java, C/C++, R
- **Software tools:** PyTorch, Scikit-learn, Pandas, Seaborn, Linux, Unix, Git, HuggingFace